

Swiggy Analytics

A PROJECT REPORT

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ABSTRACT

The Web Analytics Dashboard project is developed to analyze and visualize the digital performance of Swiggy. In the modern digital environment, businesses generate large volumes of data, making it essential to analyze this information to understand user behavior, traffic patterns, and business growth. This project focuses on transforming such data into meaningful insights through an interactive and visually appealing dashboard.

The system is designed as a web-based application that presents key performance indicators such as website traffic sources, audience demographics, financial performance, and competitive analysis. The dashboard enables users to easily interpret complex data using graphical representations like bar charts, line graphs, and doughnut charts. This helps in identifying trends, patterns, and areas for improvement.

The frontend of the system is developed using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface. The Chart.js library is used for data visualization, allowing dynamic and interactive chart generation. The system is structured into multiple modules, including traffic analysis, audience analysis, financial analysis, competitive analysis, and recommendation generation.

The results of the project show that Swiggy has strong brand recognition with a high percentage of direct traffic and consistent user engagement. The analysis also highlights potential areas for improvement, such as increasing social media presence and optimizing marketing strategies. Financial insights indicate steady growth and a movement towards profitability, while competitive analysis provides an understanding of Swiggy's position in the market.

Overall, the project demonstrates the importance of web analytics in decision-making and provides a practical implementation of data visualization techniques. The system successfully converts raw data into actionable insights, making it useful for both academic purposes and real-world applications.

CHAPTER 1: PROJECT OVERVIEW

The Web Analytics Dashboard project is developed to provide a comprehensive analysis of the digital performance of Swiggy. The project focuses on collecting, analyzing, and visualizing important data related to website traffic, user behavior, and business growth in a structured and interactive manner.

The main purpose of this project is to convert complex and large amounts of data into meaningful insights that can be easily understood through graphical representation. The system uses charts and visual elements to present key performance indicators such as traffic sources, audience demographics, financial trends, and competitive positioning. This helps users quickly identify patterns, trends, and areas that require improvement.

The project is implemented as a web-based dashboard using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface. The Chart.js library is used to create dynamic charts such as bar graphs, line charts, and doughnut charts. These visualizations enhance the understanding of data and improve decision-making capabilities.

The dashboard is divided into multiple sections, each focusing on a specific aspect of analysis. The traffic analysis section shows how users reach the platform, while the audience analysis section provides insights into user demographics and behavior. The financial analysis section highlights revenue growth and performance trends, and the competitive analysis section compares Swiggy with other platforms in the market.

The project also includes a recommendation section that provides strategic suggestions based on the insights generated. These recommendations aim to improve marketing strategies, increase user engagement, and support business growth.

Overall, this project demonstrates how web analytics can be effectively used to understand digital performance and support data-driven decision-making. It serves as a practical example of combining data analysis with modern web technologies to create an interactive and informative system.

CHAPTER 3: INTRODUCTION

In the digital era, businesses heavily depend on web analytics to understand user behavior, optimize performance, and make data-driven decisions. Web analytics involves the collection, measurement, and analysis of website data to improve user experience and achieve business goals. This project focuses on analyzing the digital performance of Swiggy, one of India's leading online food delivery platforms.

Swiggy operates in a highly competitive market where user engagement, traffic acquisition, and revenue growth are critical for success. With millions of users accessing the platform daily, it becomes essential to study how users interact with the website, where the traffic comes from, and how effectively the platform converts visitors into customers. This project aims to provide a comprehensive analysis of these aspects using a web-based analytics dashboard.

The system developed in this project is designed to present key performance indicators (KPIs) such as website traffic sources, user demographics, financial growth, and competitive analysis in a visually appealing and easy-to-understand format. The dashboard uses charts and graphs to represent data, enabling users to quickly interpret insights and identify trends.

The primary objectives of this project include analyzing traffic sources such as direct, organic, referral, and social traffic; understanding audience behavior based on age, gender, and usage patterns; evaluating financial performance including revenue growth and order value; and comparing Swiggy with its competitors in the market. By achieving these objectives, the project provides valuable insights into the strengths and areas of improvement for the platform.

The frontend of the project is developed using HTML, CSS, and JavaScript, ensuring a responsive and interactive user interface. Chart.js is used for data visualization, allowing dynamic representation of analytics data through various chart types such as bar charts, line graphs, and doughnut charts. The system is designed to be user-friendly, responsive across devices, and efficient in displaying complex data.

The motivation behind this project is to demonstrate the importance of web analytics in modern businesses and to provide a practical implementation of data visualization techniques. By converting raw data into meaningful insights, the system helps in understanding business performance and supports strategic decision-making.

CHAPTER 4: BACKGROUND STUDY

In today's digital ecosystem, web analytics has become an essential component for understanding user behavior and improving business performance. Organizations rely on analytics to track user interactions, monitor website performance, and make informed strategic decisions. This project focuses on analyzing the digital performance of Swiggy by studying its traffic patterns, audience behavior, and financial growth.

Traditionally, businesses relied on manual methods and basic statistical tools to analyze data. These approaches were time-consuming, less accurate, and not capable of handling large-scale data generated by modern web platforms. With the rapid growth of internet usage, platforms like Swiggy generate massive volumes of data daily, including user visits, clicks, transactions, and engagement metrics. This has led to the development of advanced web analytics tools and technologies.

Several existing tools such as Google Analytics, SimilarWeb, and SEMrush are widely used to analyze website performance. These tools provide insights into key metrics such as:

- Traffic sources (Direct, Organic, Referral, Social)
- User demographics (age, gender, location)
- Engagement metrics (session duration, bounce rate)
- Conversion rates and user behavior

However, these tools often present data in a generic format and may not provide customized visualizations tailored to specific project requirements. Additionally, they may require subscriptions or advanced configurations, making them less accessible for academic purposes. This project addresses these limitations by creating a customized web analytics dashboard that simplifies data interpretation through interactive visualizations.

The concept of web analytics is closely related to data science and business intelligence. It involves collecting raw data, processing it, and transforming it into meaningful insights using visualization techniques. Charts such as bar graphs, line charts, and pie charts help in identifying patterns, trends, and anomalies in the data. Visualization plays a crucial role in making complex data easy to understand and actionable.

Another important aspect of web analytics is user segmentation. By categorizing users based on age, gender, and behavior, businesses can better understand their target audience and tailor their services accordingly. In the case of Swiggy, the majority of users belong to the 25–34 age group, indicating a strong presence among young professionals. Such insights help businesses design targeted marketing strategies.

The competitive landscape is also an important part of background study. Swiggy competes with platforms like Zomato, Blinkit, and Zepto. Each competitor offers similar services but differs in strategy, user experience, and market positioning. Understanding competitor performance helps in identifying strengths and weaknesses, enabling better strategic planning.

With the advancement of web technologies, modern dashboards are built using tools like HTML, CSS, JavaScript, and libraries such as Chart.js. These technologies allow developers to create responsive, interactive, and visually appealing interfaces that enhance user experience. This

CHAPTER 5: CHALLENGES

During the development of the Web Analytics Dashboard for analyzing the digital performance of Swiggy, several challenges were encountered. These challenges were related to data handling, visualization, design, and implementation. Overcoming these issues helped in improving the overall quality and functionality of the project.

One of the major challenges was handling and organizing data from different sources. Since the project uses analytical and estimated data instead of real-time data, it was difficult to ensure accuracy and consistency across all sections. Proper structuring of data was required before it could be used for visualization.

Another challenge was designing effective data visualizations. Representing complex data in a simple and understandable format required careful selection of chart types such as bar charts, line graphs, and doughnut charts. Ensuring that each chart clearly conveyed the intended information without confusion was a critical task.

Maintaining responsiveness across different devices was also a challenge. The dashboard needed to function smoothly on mobile phones, tablets, and desktops. Adjusting layouts, charts, and UI elements to fit different screen sizes required additional effort in CSS design and testing.

Performance optimization was another important concern. Rendering multiple charts and animations on a single page could slow down the system. Therefore, it was necessary to optimize the code and ensure smooth loading and interaction without lag.

Ensuring a user-friendly interface was also challenging. The dashboard had to be simple enough for non-technical users while still providing detailed analytical insights. Balancing simplicity and functionality required careful design decisions.

Another limitation faced during the project was the absence of backend integration. Since the system is frontend-based, it does not support real-time data updates or database connectivity. This restricted the dynamic capabilities of the dashboard.

Finally, understanding and interpreting analytics data itself was a challenge. Converting raw numerical data into meaningful insights required proper analysis and domain understanding of web analytics concepts.

CHAPTER 6: DESIGN AND PROCESS

This chapter explains the overall system design, architecture, and methodology used in developing the Web Analytics Dashboard for analyzing the digital performance of Swiggy. The system is designed to provide an interactive and user-friendly interface that simplifies complex analytical data through visual representation. The main focus of the design is to ensure clarity, responsiveness, and efficient data presentation so that users can easily interpret insights.

6.1 System Architecture

The project follows a frontend-based architecture in which all analytics data is displayed through a web interface. The system is divided into three main layers that work together to ensure smooth functionality.

The frontend layer is developed using HTML, CSS, and JavaScript. It is responsible for designing the user interface and handling user interactions. The interface is designed to be responsive so that it works efficiently on different devices such as mobile phones, tablets, and desktops.

The visualization layer is implemented using the Chart.js library. This layer converts raw data into graphical formats such as bar charts, line graphs, and doughnut charts. These visual elements make it easier for users to understand patterns and trends in the data.

The data layer consists of structured analytical data collected from various sources. This data includes important metrics such as website traffic, user behavior, and financial performance. The data is processed and then passed to the visualization layer for graphical representation.

This layered architecture ensures a smooth flow from data processing to visualization and finally to user interaction, making the system efficient, scalable, and easy to use.

6.2 System Modules

The system is divided into different modules, each responsible for analyzing a specific aspect of Swiggy's performance. These modules work together to provide a complete understanding of the platform's digital analytics.

The Traffic Analysis Module focuses on identifying where users are coming from. It analyzes different traffic sources such as direct visits, organic search, referral links, and social media. This helps in understanding how effective the platform's marketing strategies are and which channels are performing better.

The Audience Analysis Module provides insights into user demographics and behavior. It includes information such as age distribution, gender ratio, and user engagement patterns. This module helps in identifying the target audience and understanding how users interact with the platform.

The Financial Analysis Module evaluates the business performance of Swiggy. It analyzes revenue growth, net loss trends, and gross order value. This module helps in understanding the financial health of the company and its growth over time.

The Competitive Analysis Module compares Swiggy with its competitors such as Zomato, Blinkit, and Zepto. It highlights differences in traffic, services, and market positioning, which helps in identifying strengths and weaknesses.

The Recommendation Module provides strategic suggestions based on the analysis. It focuses on improving marketing strategies, enhancing user engagement, and identifying growth

opportunities for the platform.

6.3 Process Flow

The system follows a structured process to convert raw data into meaningful insights.

First, data is collected from analytical sources such as reports and online tools. This data includes information about traffic, users, and financial performance.

Next, the collected data is processed and organized into a structured format so that it can be easily used for analysis.

After processing, the data is visualized using Chart.js. Different types of charts are used to represent different types of data, making it easier to understand patterns and trends.

Finally, insights are generated based on the visualized data. These insights help in understanding performance and making better decisions.

6.4 Design Considerations

While designing the system, several important factors were taken into account to ensure its effectiveness.

The system is designed with a user-friendly interface so that users can easily navigate and understand the dashboard without any confusion. A clean and simple layout is used to improve readability.

Responsiveness is another key factor, ensuring that the system works smoothly on all types of devices, including mobile phones and tablets.

Performance is optimized so that charts load quickly and animations run smoothly without lag. The system is also designed to be scalable, meaning it can be extended in the future to include real-time data and advanced analytics features.

Accuracy is maintained by ensuring that all data is correctly represented and visualized without any distortion.

6.5 Tools and Technologies Used

The project uses modern web technologies to build an efficient and interactive dashboard.

HTML is used to create the structure of the webpage, while CSS is used for styling and layout design. JavaScript is used to add dynamic functionality and interactivity. The Chart.js library is used to create charts and graphs for data visualization.

6.6 Advantages of the Design

The system provides an interactive and visually appealing dashboard that makes it easy to understand complex analytics data. It helps users quickly interpret insights and supports better decision-making. The design ensures efficient data representation and a smooth user experience.

6.7 Limitations of the System

Despite its advantages, the system has some limitations. It uses static data instead of real-time

CHAPTER 7: RESULT ANALYSIS AND VALIDATION

This chapter presents the evaluation of the Web Analytics Dashboard developed for analyzing the digital performance of Swiggy. The purpose of this analysis is to determine how effectively the system meets its objectives and how accurately it represents analytical data through visualizations. The results are interpreted based on traffic patterns, audience behavior, financial performance, and competitive positioning.

7.1 Result Analysis

The analysis of the system reveals several important insights about Swiggy's digital presence. The dashboard successfully transforms raw data into meaningful visual information, making it easier to understand trends and patterns.

The traffic analysis shows that a major portion of users comes from direct traffic, accounting for approximately 56.9%. This indicates strong brand recognition and user loyalty. Organic search contributes around 28.2%, which reflects good search engine visibility. However, social media and referral traffic are comparatively lower, suggesting potential areas for marketing improvement.

The audience analysis highlights that the majority of users belong to the 25–34 age group, indicating that young professionals form the core user base. The gender distribution shows a higher percentage of male users, approximately 66%, while female users account for the remaining portion. The data also indicates a high order frequency, suggesting strong user engagement and repeated usage of the platform.

From a financial perspective, the system shows a consistent growth trend in revenue, with an increase of around 36% year-over-year. At the same time, net losses are gradually decreasing, indicating that the company is moving towards profitability. The Gross Order Value (GOV) has also increased, reflecting higher transaction volumes on the platform.

The competitive analysis demonstrates that Swiggy operates in a highly competitive environment with major competitors such as Zomato, Blinkit, and Zepto. Among these, Zomato appears to have a higher traffic volume, but Swiggy maintains a strong position due to its multi-service platform that includes food delivery, grocery services, and dining options.

7.2 Validation of the System

Validation ensures that the system works correctly and produces accurate results under different conditions. The system was evaluated based on several key parameters.

The accuracy of the system is maintained by ensuring that all charts correctly represent the given data without any inconsistencies. Each visualization corresponds to the actual dataset used in the project.

Efficiency is achieved through fast loading times and smooth rendering of charts. The dashboard responds quickly to user interactions, ensuring a seamless experience.

Usability is one of the major strengths of the system. The interface is simple, clean, and easy to navigate, allowing users to understand analytics without requiring technical expertise.

Reliability is ensured by testing the system under different scenarios. The system performs consistently without crashes or errors, even when handling multiple visual elements

simultaneously.

7.3 Testing Approach

The system was tested using different approaches to ensure its correctness and performance.

Functional testing was performed to verify that all modules, such as traffic analysis, audience analysis, and financial analysis, work as expected. Each feature was checked to ensure proper operation.

Performance testing was conducted to evaluate how quickly the system loads and displays charts. The results show that the system performs efficiently under normal conditions.

Usability testing focused on the user experience. The dashboard was designed to be intuitive, and users can easily navigate between different sections without confusion.

Validation testing was used to compare expected results with actual outputs. This ensures that the system provides accurate and reliable information.

7.4 Limitations Identified

Although the system performs well, certain limitations were identified during the analysis. The system uses static data instead of real-time data, which may not always reflect current trends. Additionally, the absence of backend integration limits dynamic data processing. The system also depends on external data sources, which may affect accuracy if the data changes.

7.5 Overall Evaluation

Overall, the Web Analytics Dashboard successfully achieves its objectives. It provides a clear and interactive representation of data, making it easier to analyze Swiggy's digital performance. The system demonstrates good accuracy, efficiency, usability, and reliability, making it suitable for academic and analytical purposes.

CHAPTER 8: CONCLUSION AND FUTURE WORK

8.1 Conclusion

The Web Analytics Dashboard developed for analyzing the digital performance of Swiggy successfully demonstrates how data can be transformed into meaningful insights through visualization. The system effectively integrates various analytical components such as traffic analysis, audience behavior, financial performance, and competitive comparison into a single interactive platform.

The project achieves its primary objective of simplifying complex data by presenting it in the form of charts and graphs. This makes it easier for users to understand patterns, trends, and relationships within the data. The dashboard provides a clear overview of key performance indicators, enabling users to analyze how Swiggy attracts users, how customers interact with the platform, and how the business is growing over time.

One of the major outcomes of this project is the identification of strong brand presence through high direct traffic and consistent user engagement. At the same time, it highlights areas of improvement, such as the need to enhance social media and referral traffic channels. The financial analysis indicates steady growth and a positive trend towards profitability, while the competitive analysis shows that Swiggy maintains a strong position in a highly competitive market.

The system also performs well in terms of usability, efficiency, and reliability. The interface is user-friendly and responsive, allowing smooth interaction across different devices. The use of modern web technologies ensures fast performance and effective data visualization.

Overall, the project successfully demonstrates the importance of web analytics in decision-making and provides a practical implementation of an analytics dashboard. It serves as a useful tool for understanding digital performance and can be further enhanced for real-world applications.

8.2 Future Work

Although the system meets its current objectives, there are several opportunities for improvement and expansion in the future.

The system can be enhanced by integrating real-time data using APIs, which will allow the dashboard to display live analytics instead of static data. This will make the system more accurate and useful for real-world applications.

Another important improvement is the addition of backend functionality. By integrating a backend system, the dashboard can store, process, and manage large volumes of data more efficiently. This will also enable user authentication and personalized analytics.

The project can be extended by incorporating advanced analytics techniques such as machine learning and predictive analysis. This will allow the system to forecast trends, predict user behavior, and provide smarter recommendations.

The development of a mobile application version of the dashboard can improve accessibility and allow users to access analytics on the go. Additionally, integrating the system with tools like Google Analytics can provide more detailed insights.

CHAPTER 9 : OUTPUT

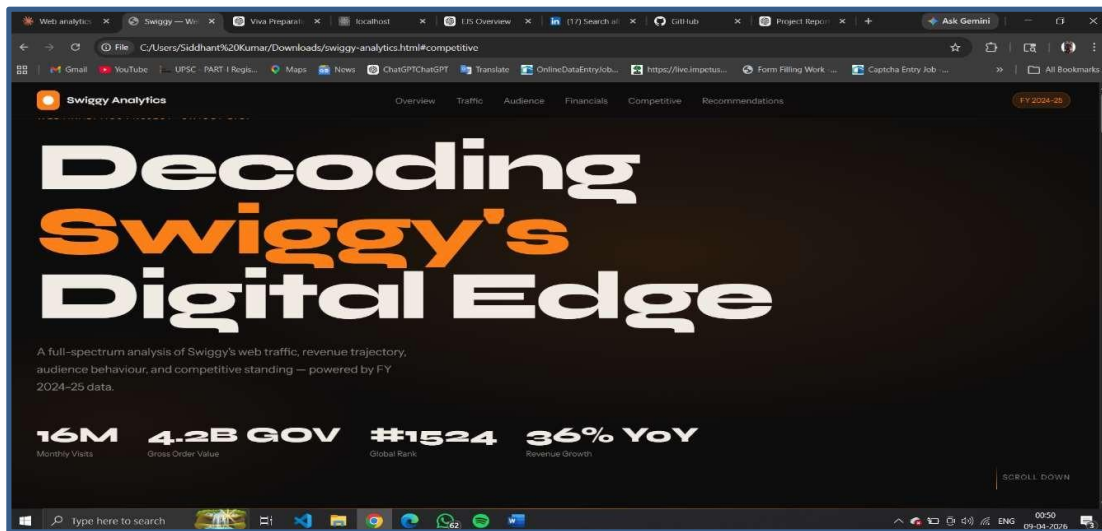


Figure 1:- Hero Section

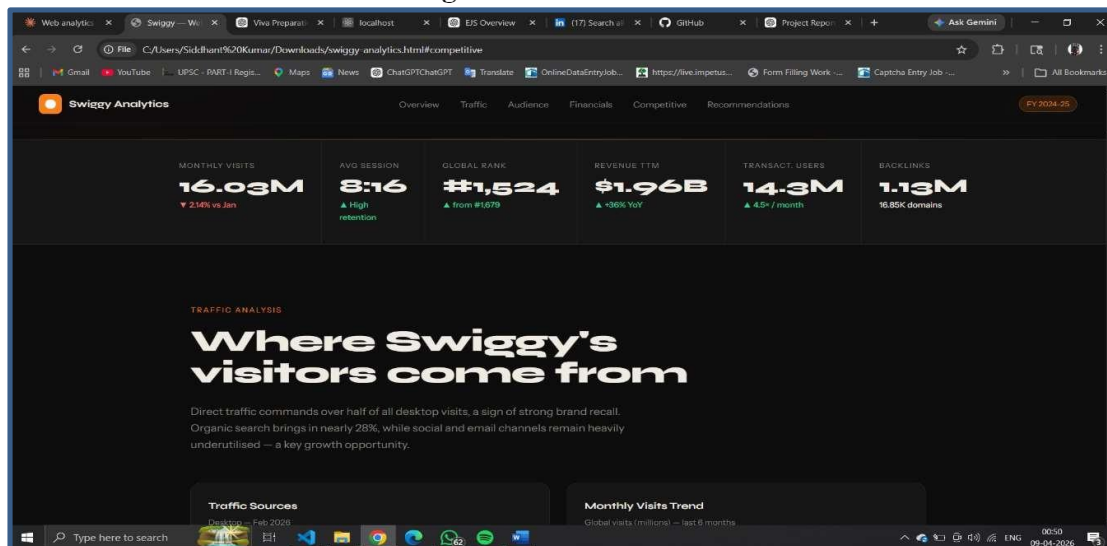


Figure 2:- Overview

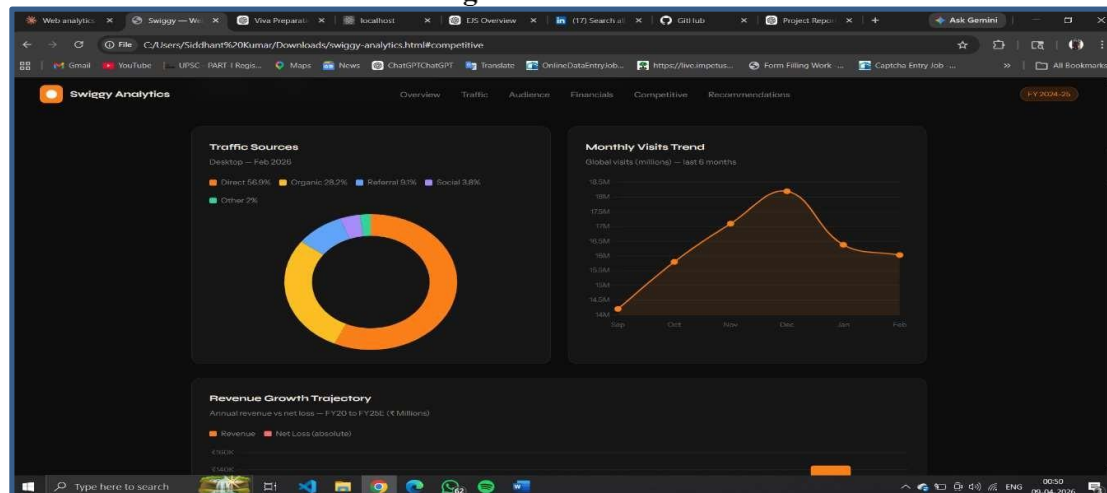


Figure 3:- Graphical Representation

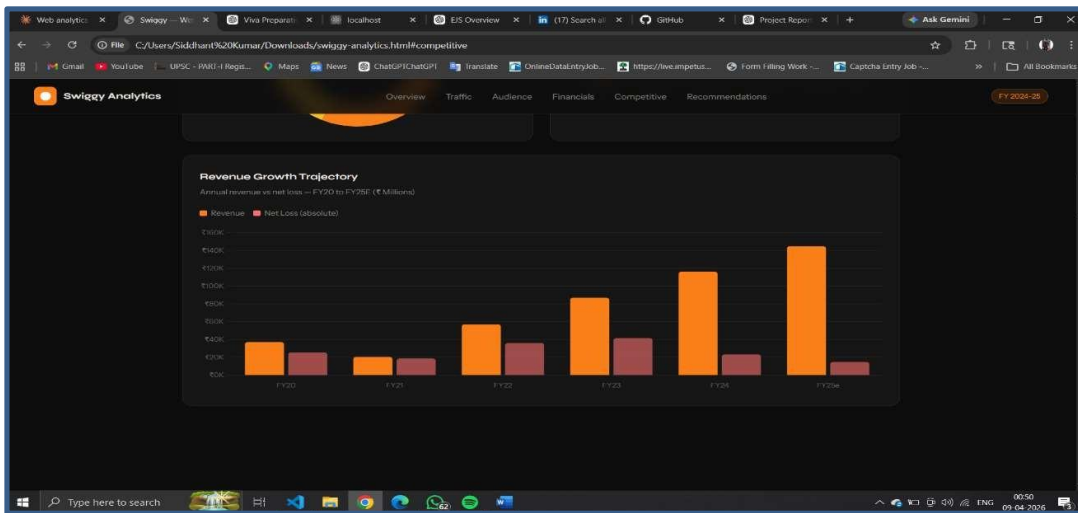


Figure 4:- Marketplace

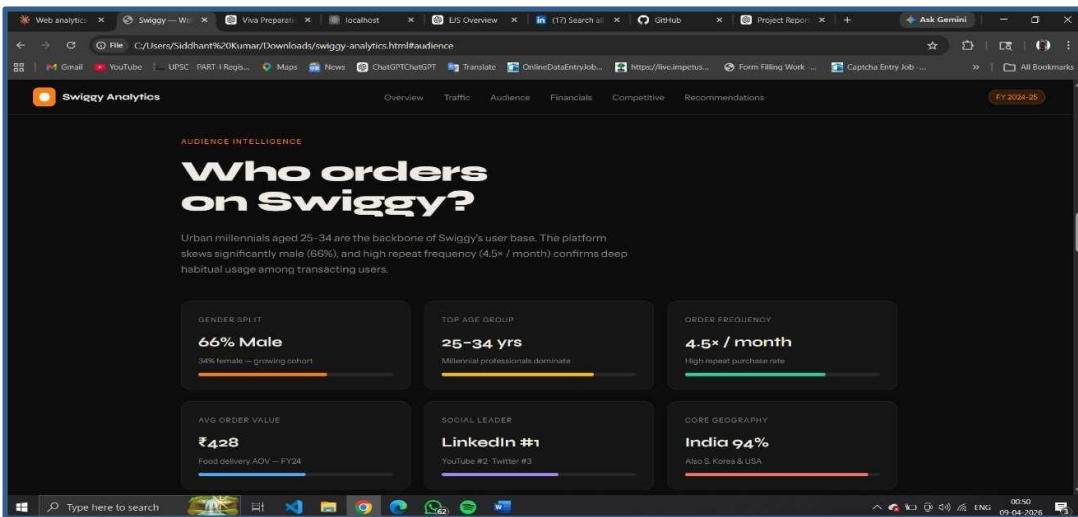


Figure 5:- Audience Intelligence

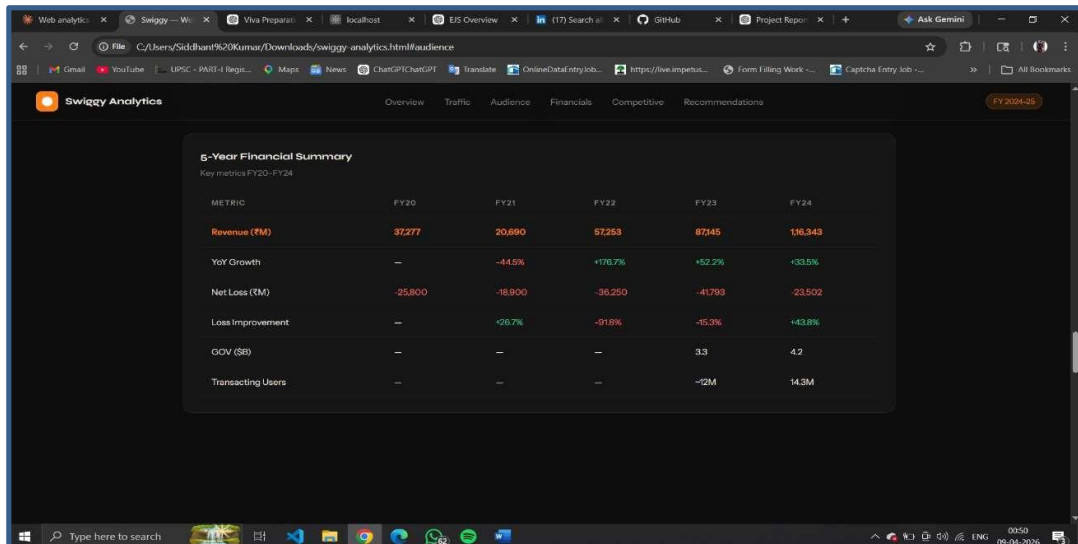


Figure 6:- Financial Summary

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